R-Hub

Enterprise Windows Infrastructure Laboratory

Project Documentation

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Bachelor of Information Technology

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Project Technologies

Windows Server 2022

Active Directory Domain Services (AD DS)

DNS Group Policy (GPO)

Organizational Units (OU)

Domain Join

GLPI IT Service Management (ITSM)

PowerShell

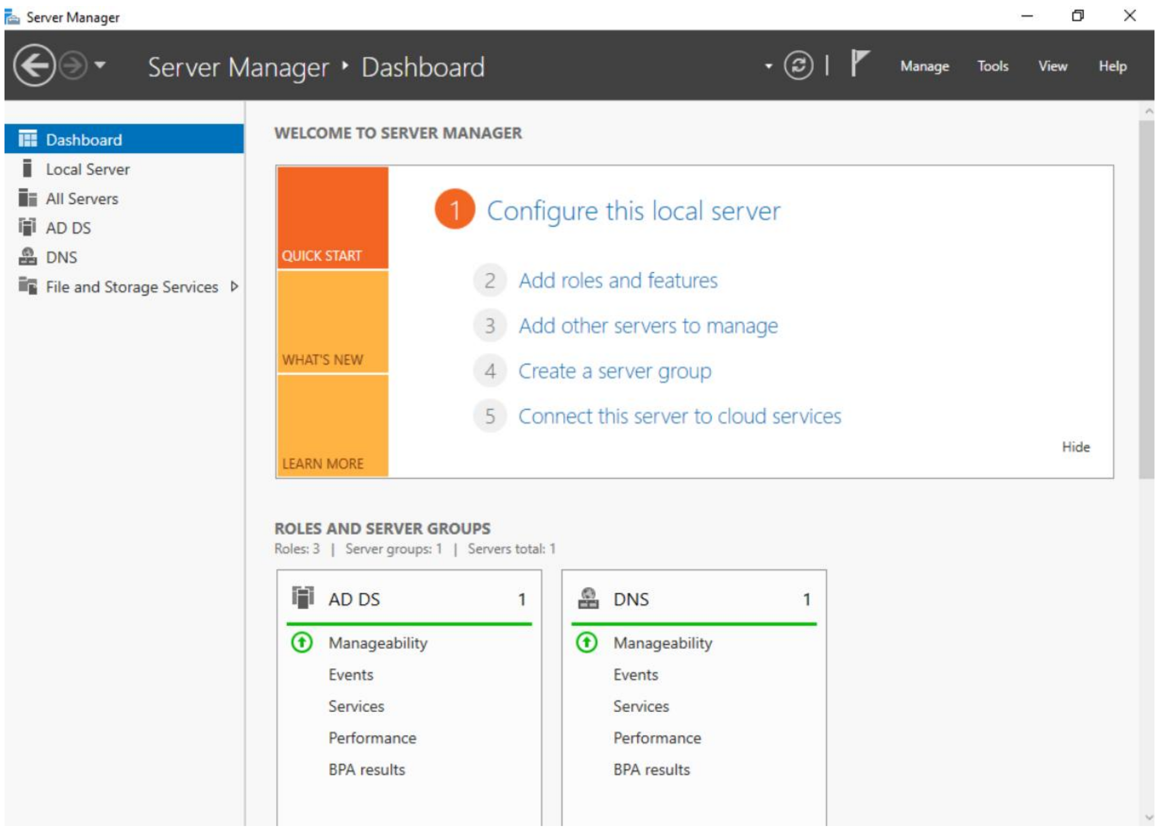
Introduction

R-Hub is a self-built enterprise IT infrastructure laboratory designed to simulate a real-world corporate environment. The project demonstrates the deployment and administration of core Microsoft infrastructure services, including Windows Server 2022, Active Directory Domain Services (AD DS), DNS, Organizational Units (OU), Group Policy (GPO), domain-joined client devices, PowerShell administration, and GLPI IT Service Management (ITSM). The objective of this project is to strengthen practical system administration skills and simulate enterprise IT operations using industry-standard technologies.

Project Objectives

- Build a Windows Server 2022 enterprise environment.
- Deploy and configure Active Directory Domain Services (AD DS).
- Configure DNS to support domain services.
- Design Organizational Units (OU) for different departments.
- Join client computers to the Active Directory domain.
- Implement Group Policy Objects (GPO).
- Deploy GLPI IT Service Management (ITSM).
- Perform administrative tasks using PowerShell.
- Simulate an enterprise IT support workflow.

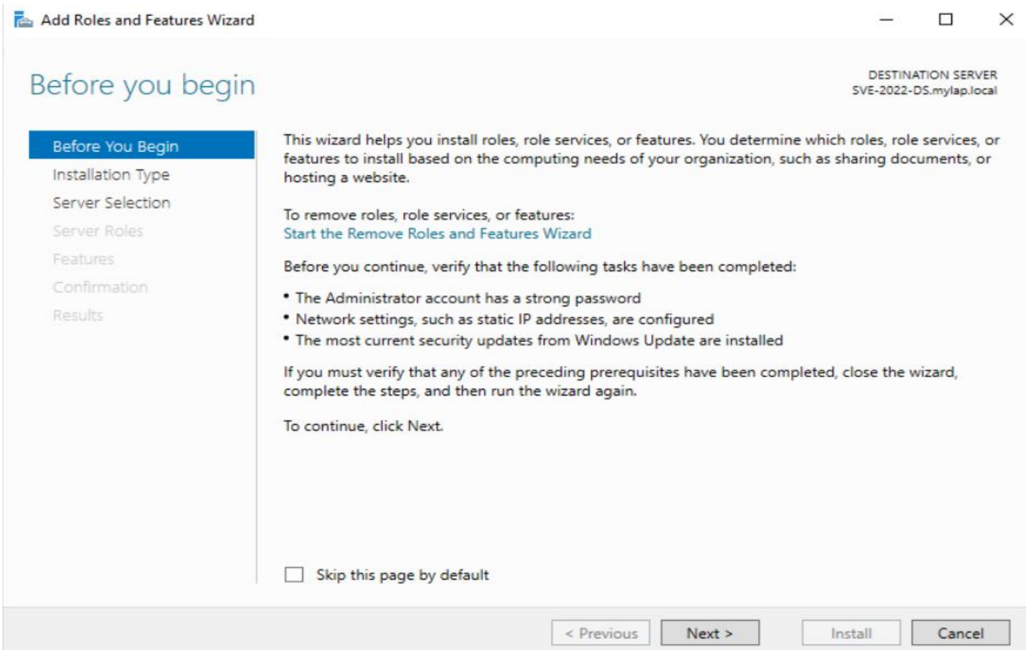
Windows Server 2022



Server Manager Dashboard

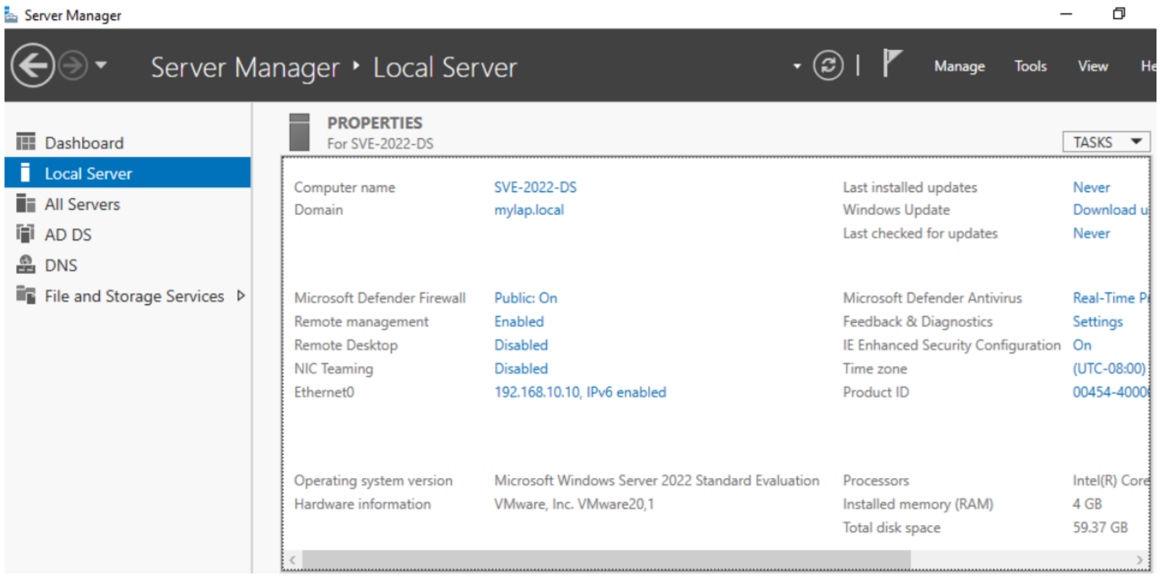
Windows Server 2022 was successfully deployed and configured as the primary server for the R-Hub environment. Server Manager was used to manage server roles and monitor installed services.

After deploying Windows Server 2022, the required infrastructure roles were installed using the Add Roles and Features Wizard



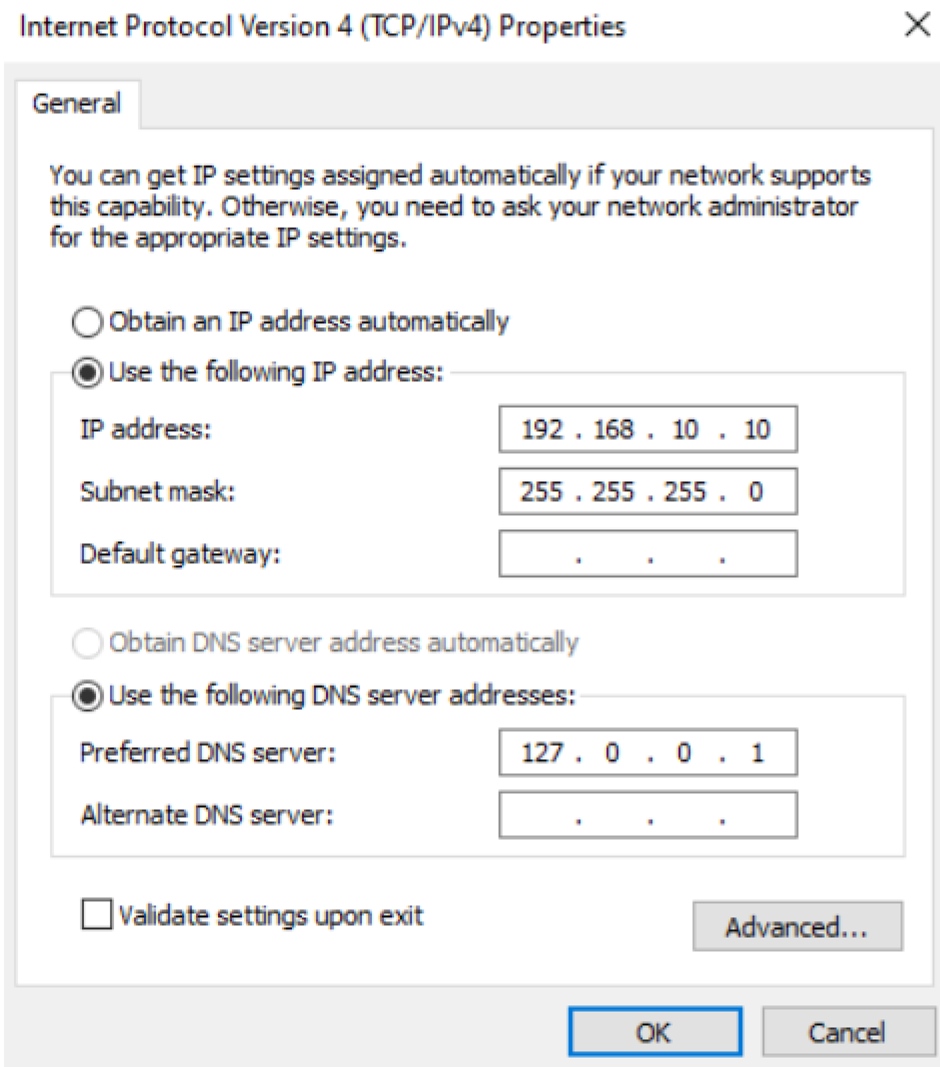
Installing Roles and Features

The Add Roles and Features Wizard was used to install the required server roles, including Active Directory Domain Services (AD DS) and DNS, which provide the foundation for the enterprise environment.



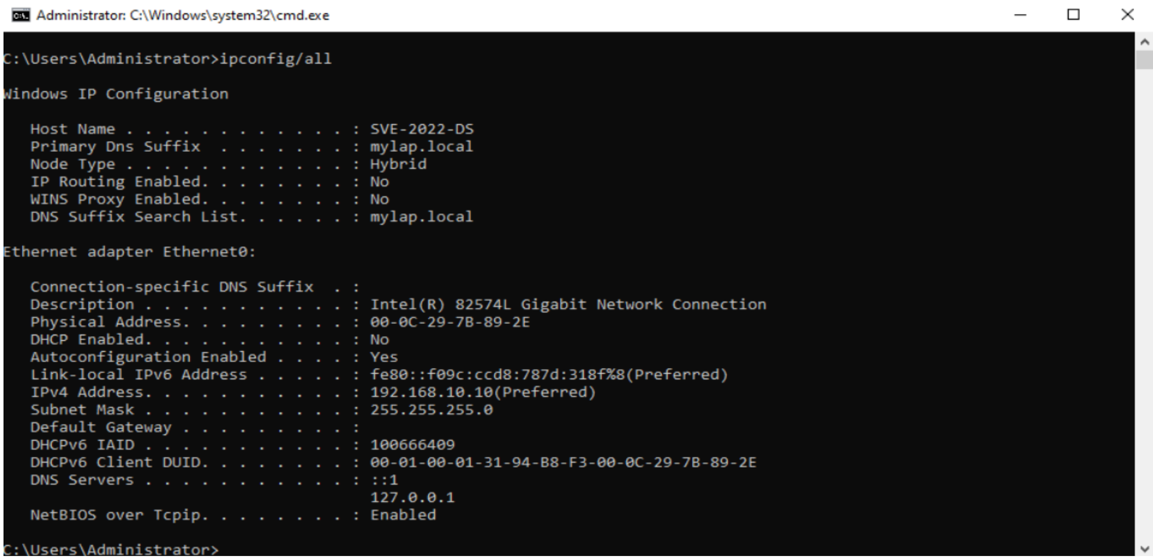
Local Server Configuration

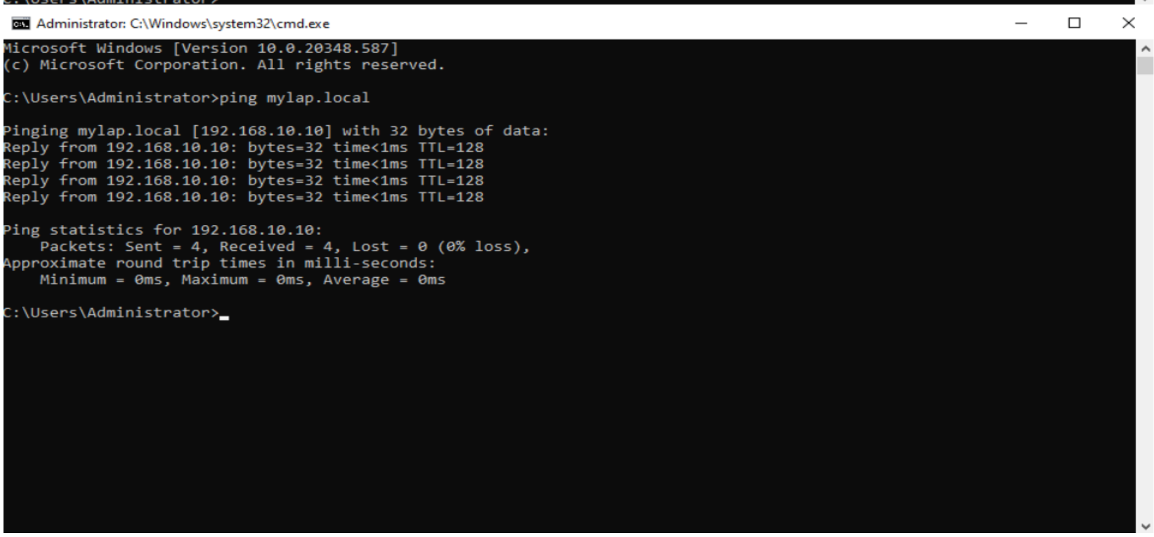
The Local Server configuration was reviewed to verify the server settings, including the computer name, domain membership, and static IP address. This ensured that the server was correctly configured before continuing with the Active Directory deployment.



Static IPv4 Configuration

A static IPv4 address was configured for the server to provide a stable network identity required for Active Directory and DNS services. The server was assigned a fixed IP address (192.168.10.10), a subnet mask of **255.255.255.0**, and the local DNS server (**127.0.0.1**) to ensure reliable name resolution and uninterrupted **Active Directory services**.

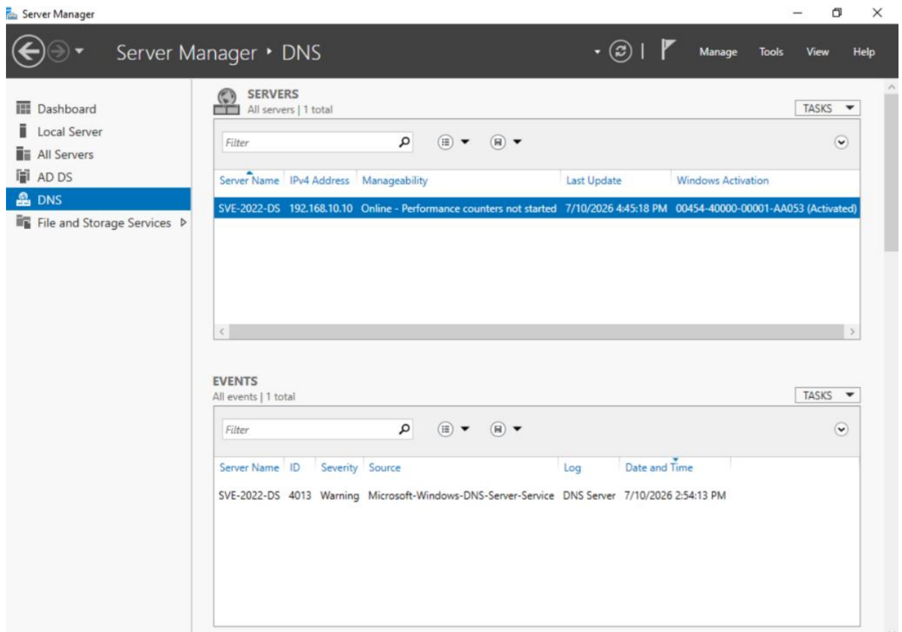


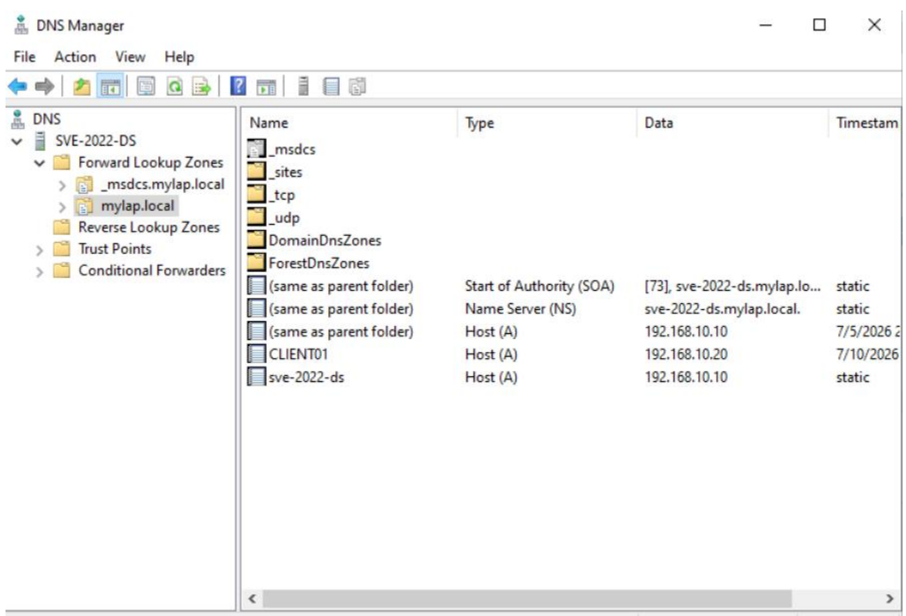


Network Configuration Verification

The network configuration was verified using the **ipconfig /all** command to confirm the assigned IP address, subnet mask, DNS settings, and domain configuration. Network connectivity was then validated by successfully pinging the server hostname (**mylap.local**), confirming successful DNS name resolution and end-to-end network connectivity within the enterprise environment.

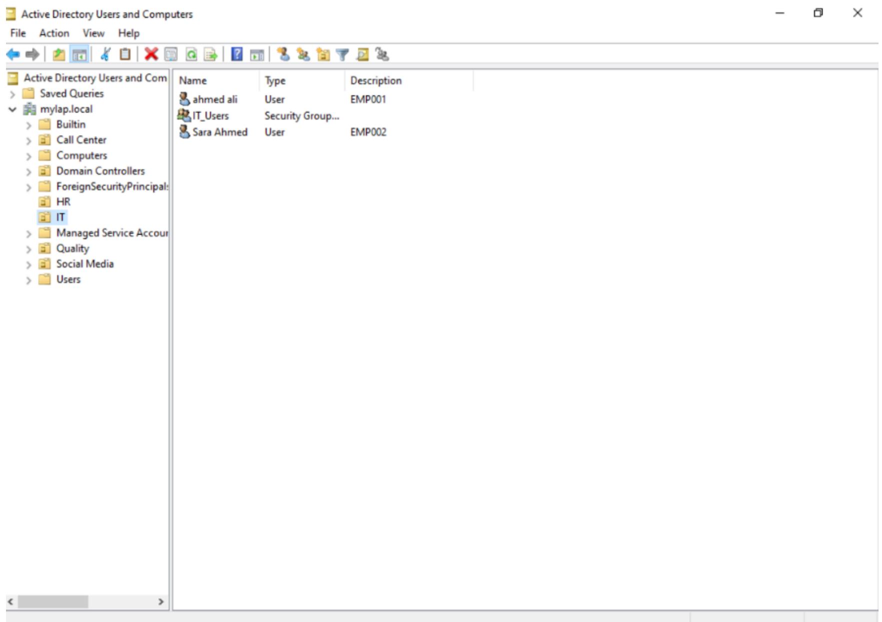
DNS Configuration and Zone Verification

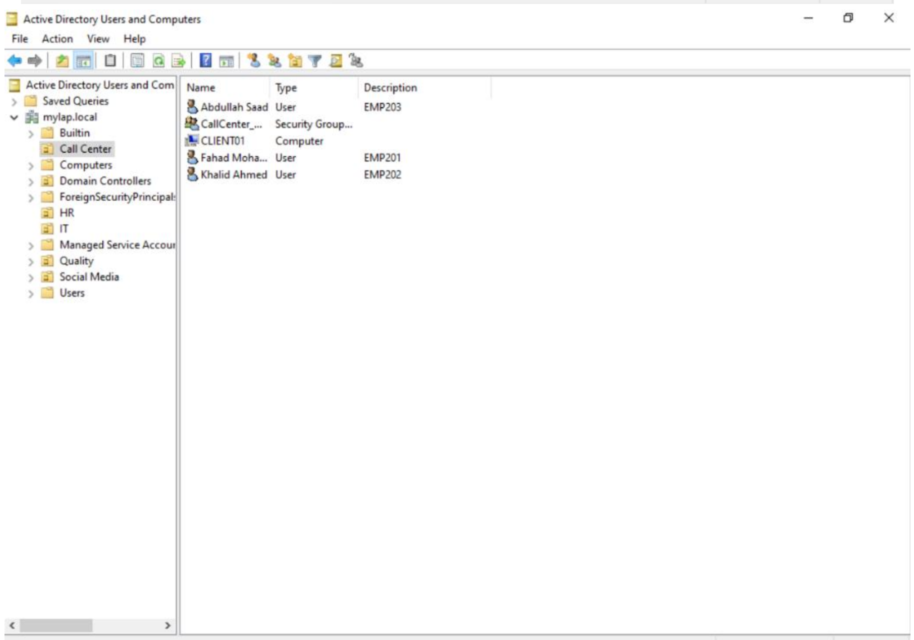




The DNS Server role was successfully configured to support the Active Directory infrastructure. The **Forward Lookup Zone (mylap.local)** was created and verified, ensuring that the required DNS records were available for reliable hostname resolution, domain authentication, and communication between domain-joined systems.

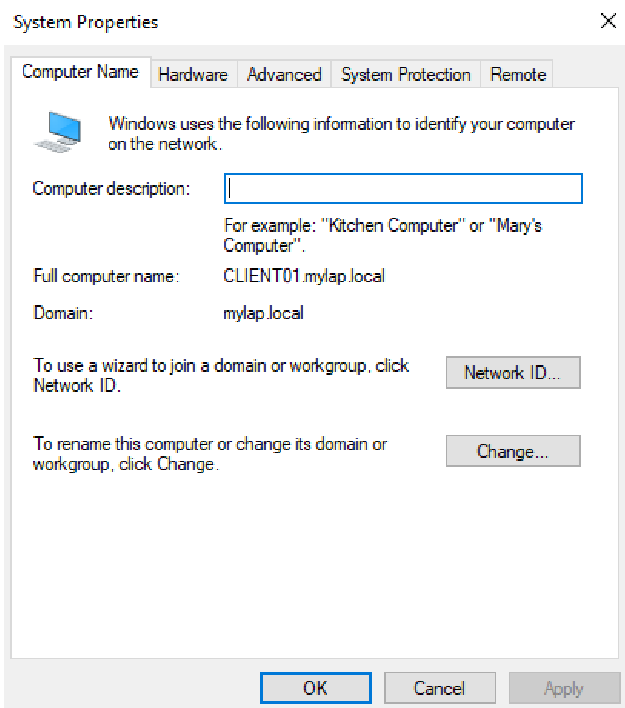
Organizational Units (OU) Structure

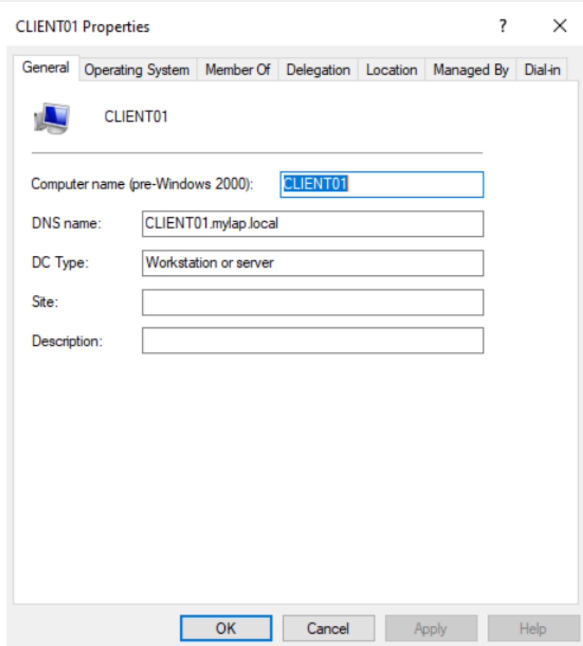




Organizational Units (OUs) were created to organize the Active Directory environment based on business departments. Separate OUs were implemented for **IT, HR, Call Center, Quality, Social Media, and Users**, providing a structured hierarchy for managing users, computers, security groups, and future Group Policy Objects (GPOs). This organizational structure simplifies administration, improves scalability, and enables department-specific policies without affecting the entire domain.

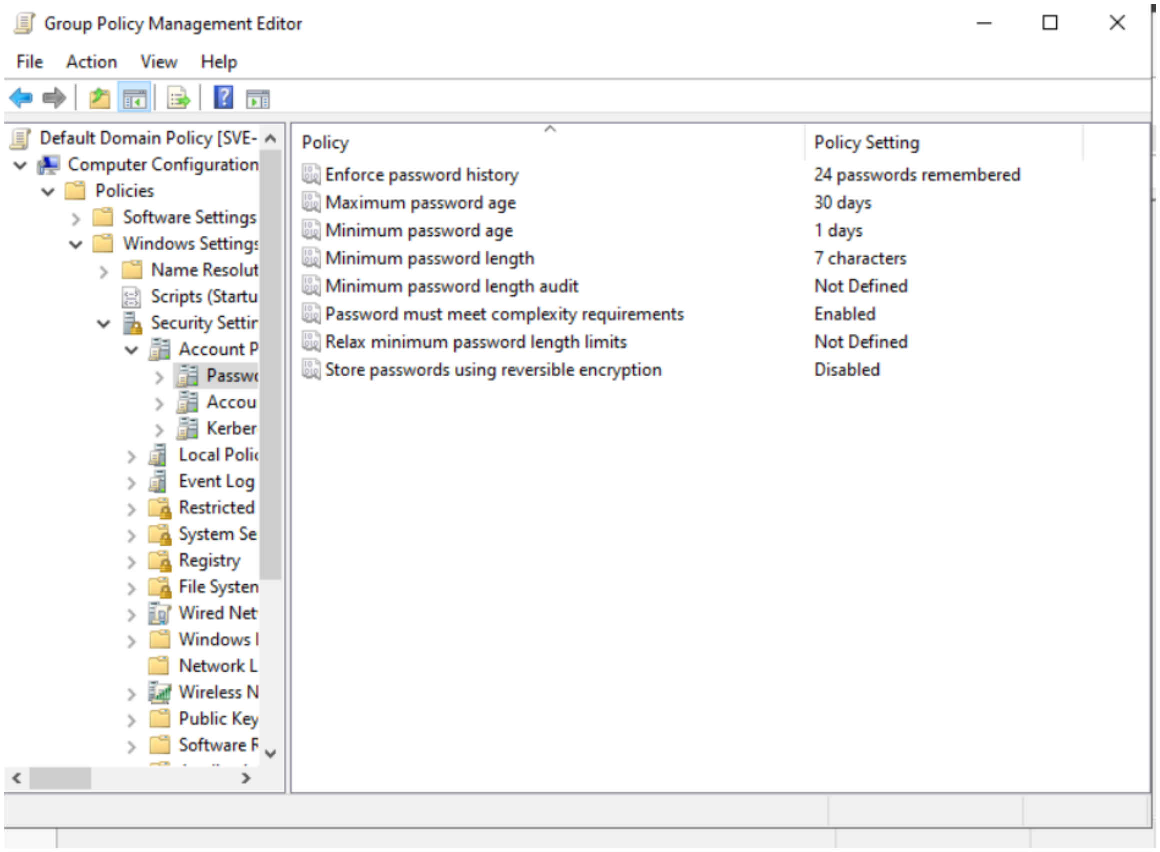
Client Domain Join Verification





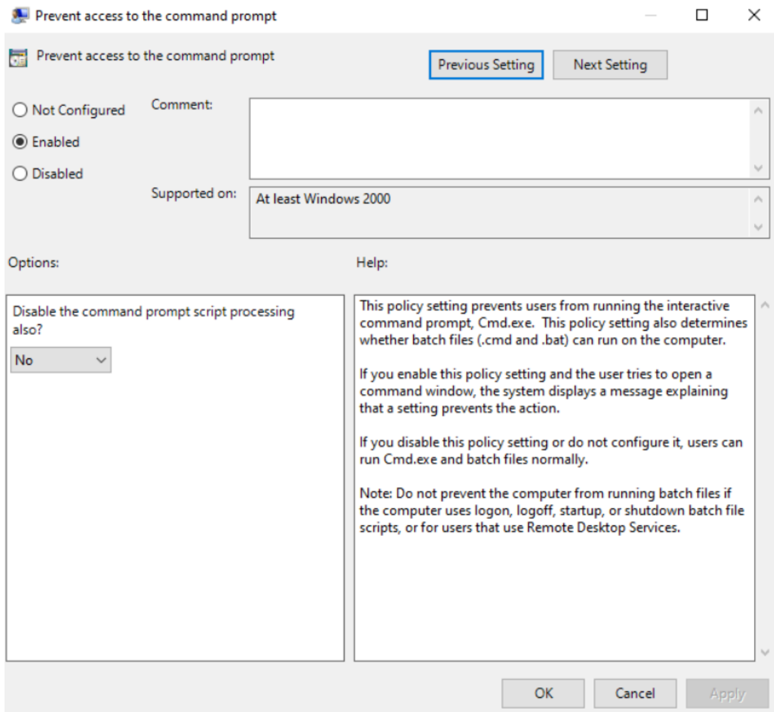
The Windows client (**CLIENT01**) was successfully joined to the **mylap.local** Active Directory domain. The first image confirms that the workstation became a member of the domain, while the second image verifies that the computer account was automatically created in Active Directory. This confirms successful communication between the client and the Domain Controller, enabling centralized authentication, **Group Policy (GPO) deployment**, centralized administration, and secure management of domain-joined devices.

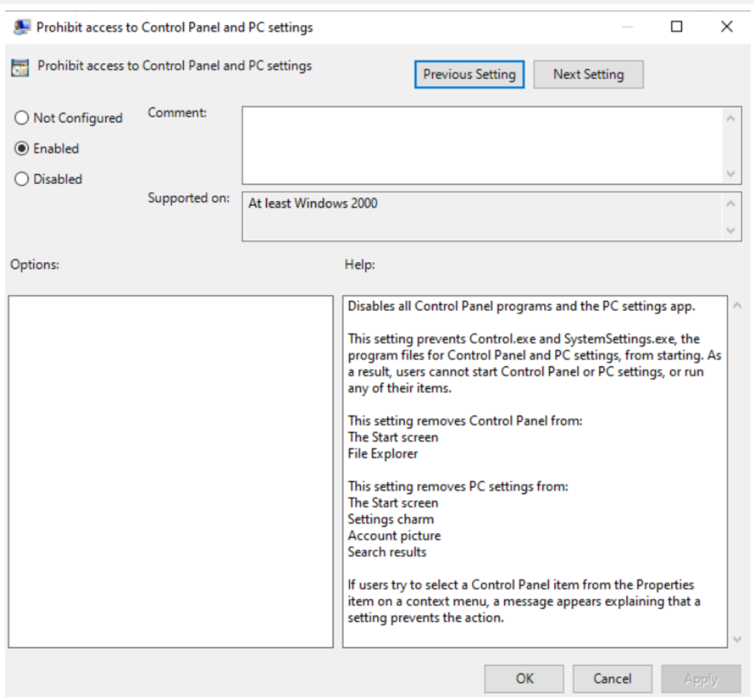
Password Policy Configuration



A domain-wide password policy was configured through Group Policy to enforce strong password requirements. The policy includes password complexity, minimum password length, and other security settings to improve account protection and reduce the risk of unauthorized access. Applying this policy at the domain level ensures consistent authentication standards across all domain-joined users and computers.

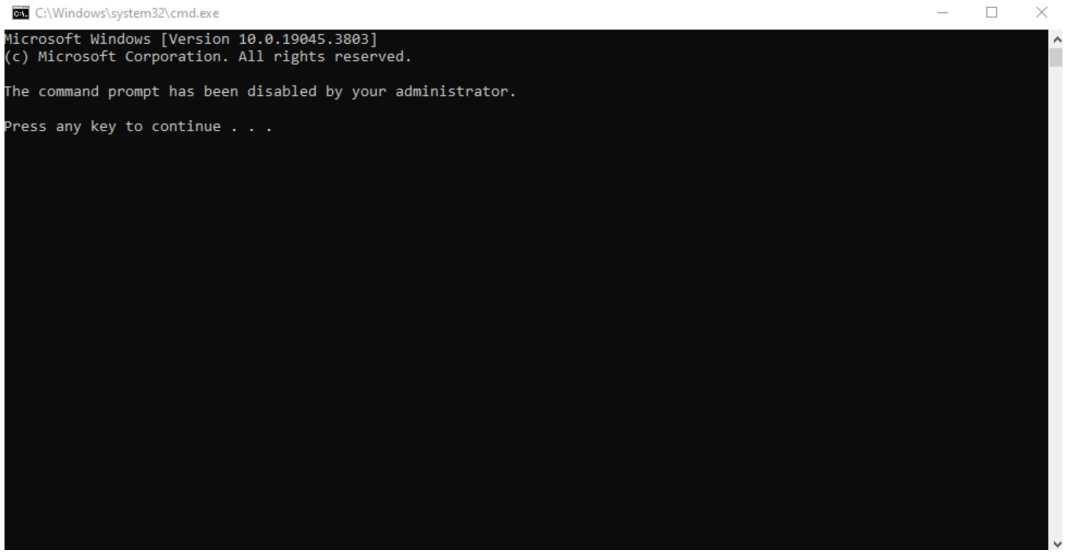
Command Prompt and Control Panel Restrictions

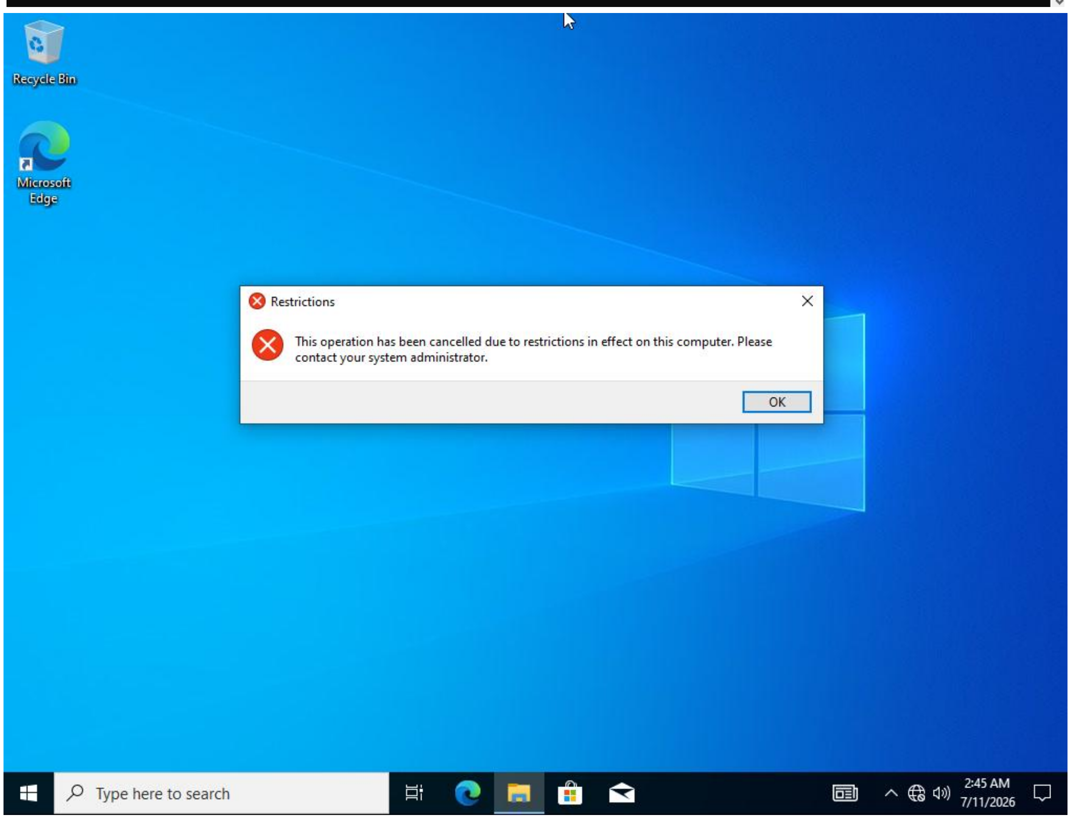




Group Policy Objects (GPOs) were configured and linked to the appropriate Organizational Units to restrict access to both the Command Prompt and the Control Panel. These policies prevent users from executing administrative commands or modifying critical system settings, helping maintain a secure, standardized, and centrally managed workstation environment across the Active Directory domain.

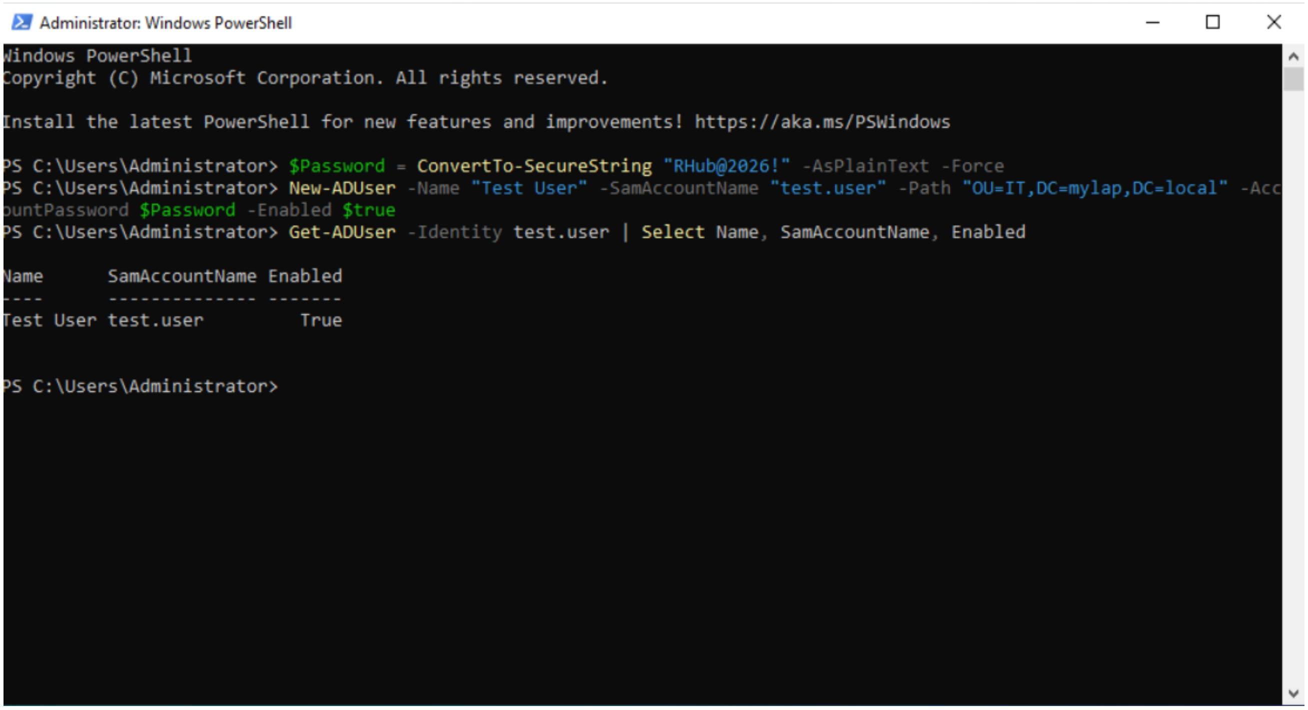
Group Policy Deployment Verification





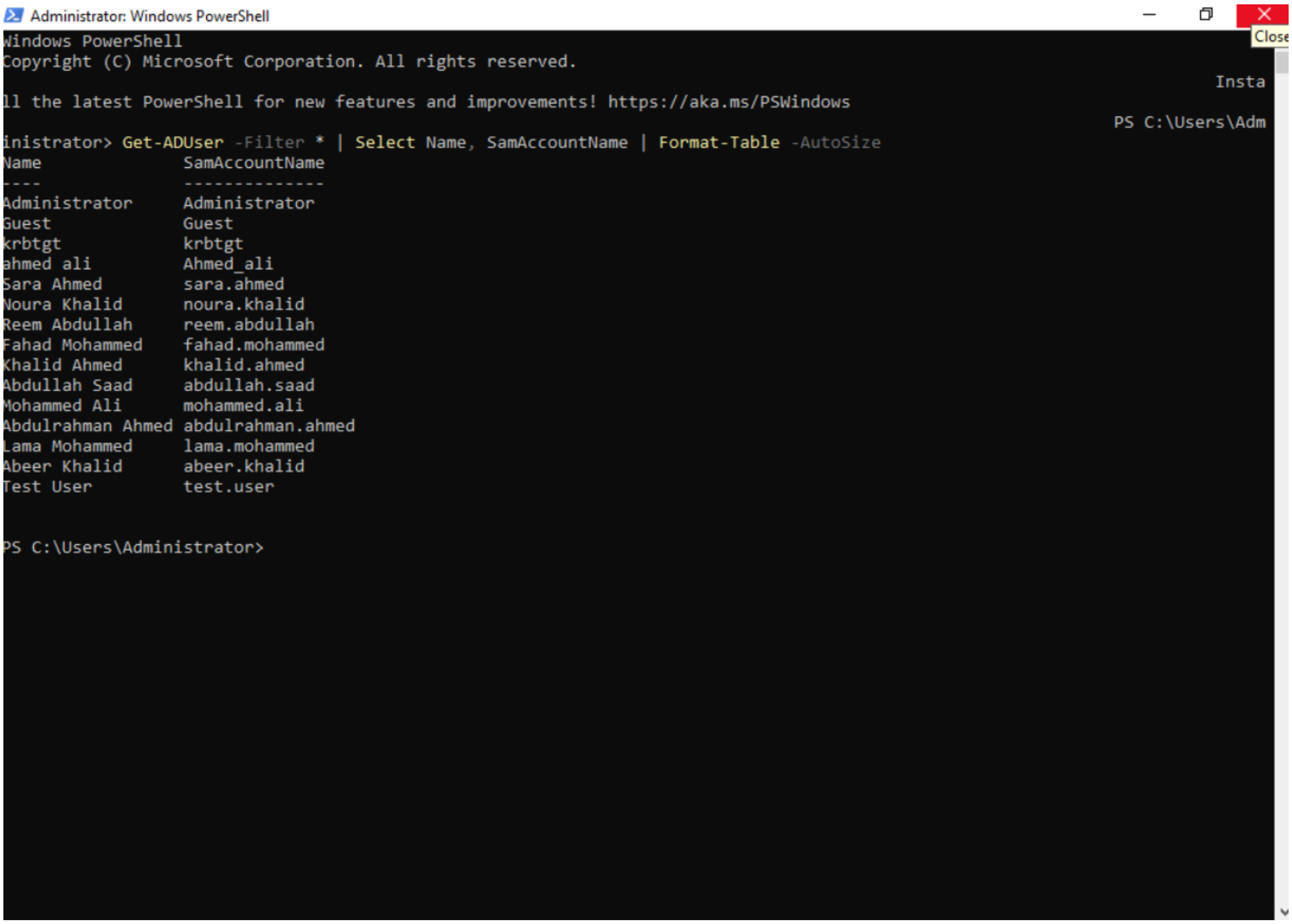
*The configured Group Policy Objects (GPOs) were successfully applied to the client workstation. The verification results confirm that access to both the **Command Prompt** and the **Control Panel** was restricted as intended, demonstrating successful policy deployment, proper Active Directory communication, centralized administration, and consistent security enforcement across **domain-joined workstations**.*

PowerShell Active Directory User Creation



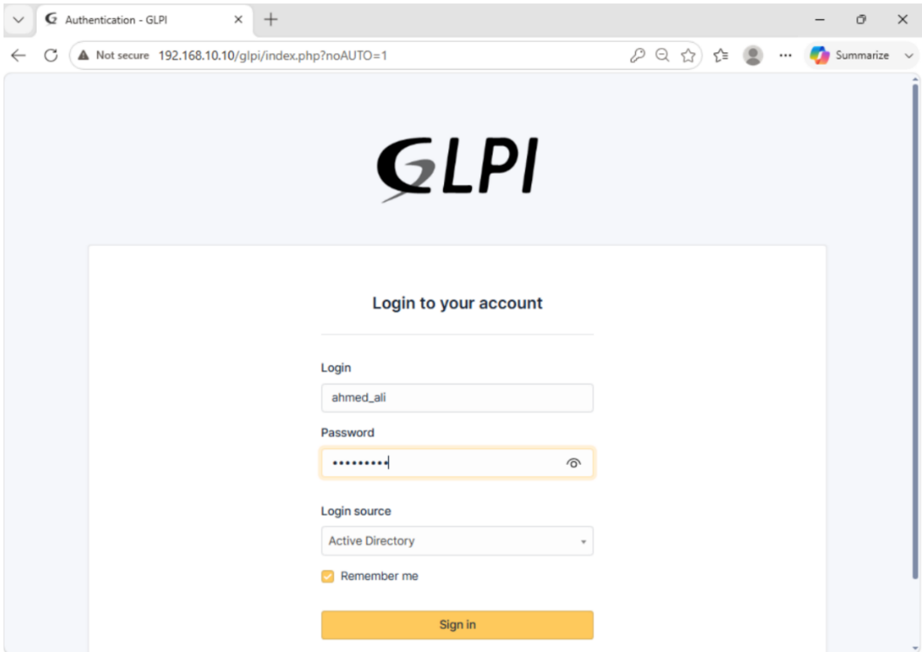
The PowerShell Active Directory module was used to automate the creation of a new domain user account. The **New-ADUser** cmdlet created the user **test.user**, and the account was successfully verified using **Get-ADUser**. This demonstrates how administrative tasks can be automated through PowerShell, reducing manual effort while ensuring consistent and efficient user provisioning across the Active Directory environment.

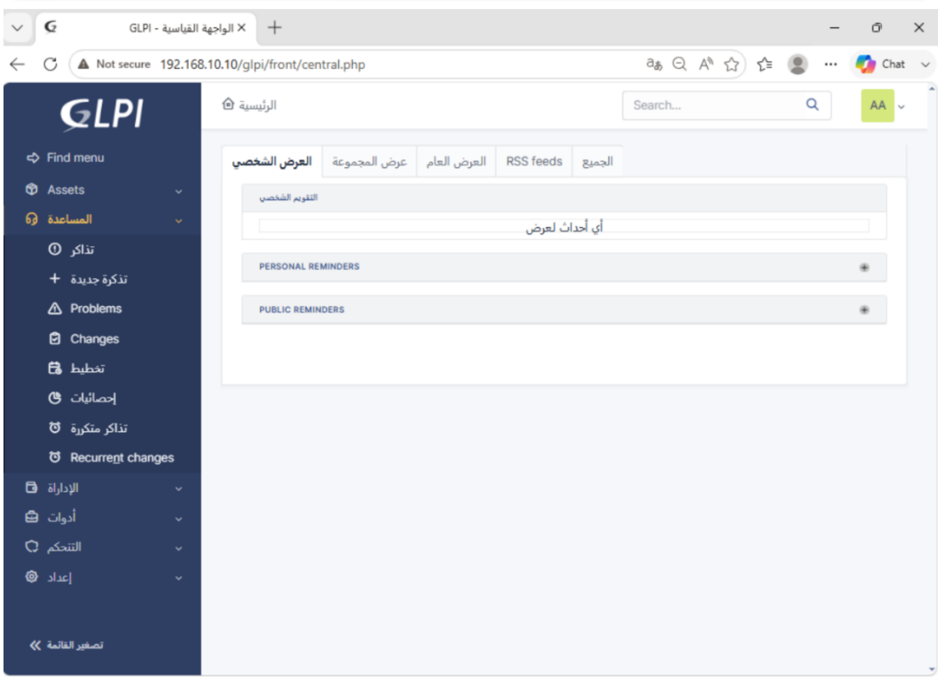
PowerShell Active Directory User Retrieval



PowerShell was used to retrieve Active Directory user accounts by executing the **Get-ADUser** cmdlet. The output displays all available domain user accounts, **confirming successful retrieval of Active Directory user information** and demonstrating efficient directory management, simplified administration, and the ability to automate user inventory and reporting tasks through PowerShell.

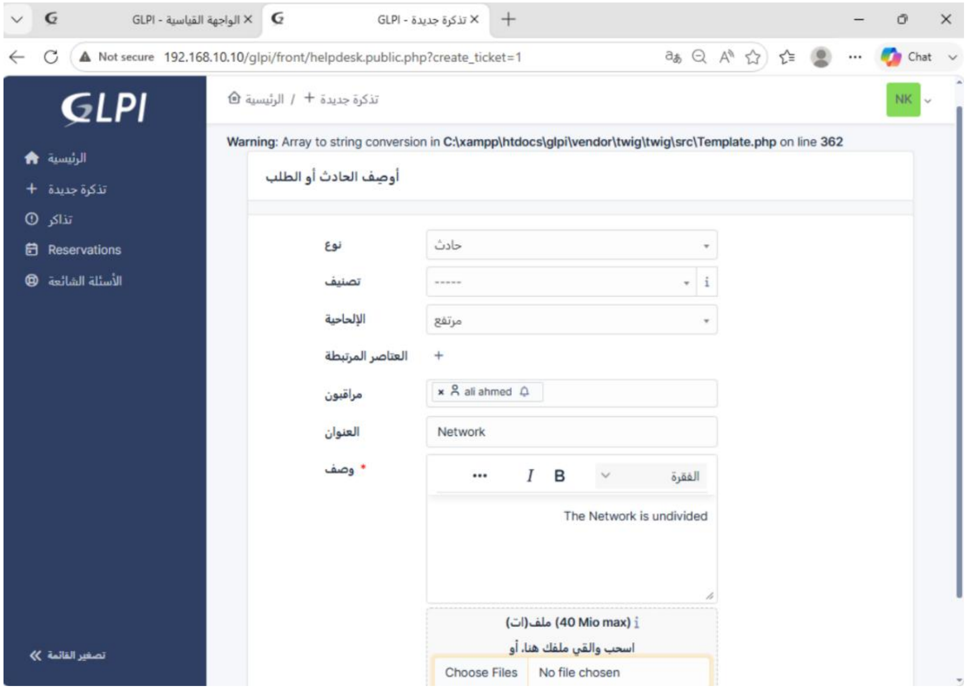
GLPI Authentication and Dashboard

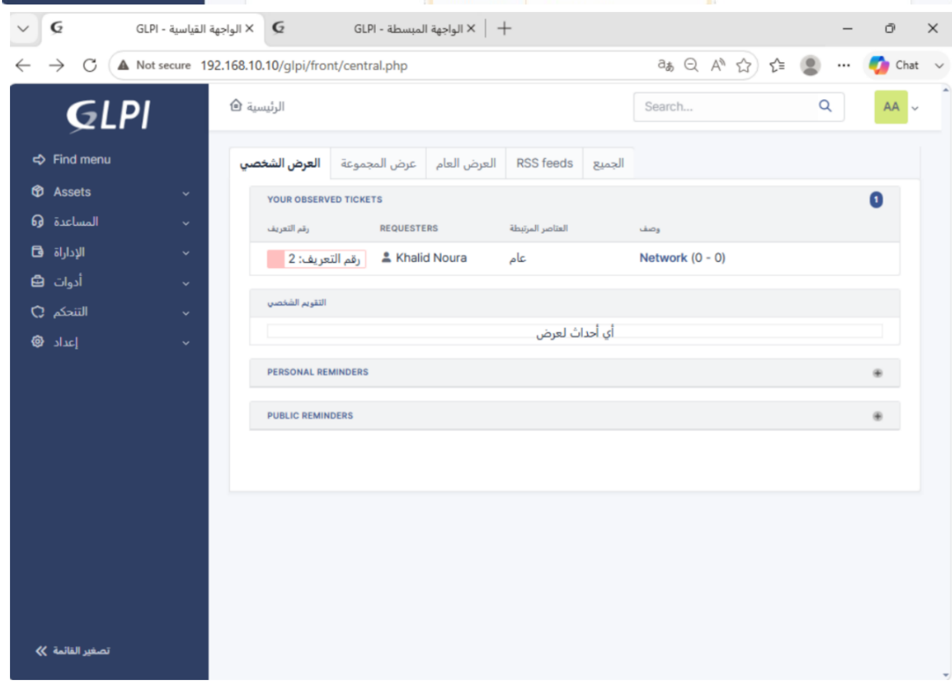




The GLPI platform was successfully accessed through the web interface using an authorized administrator account. After authentication, the dashboard provided an overview of the IT Service Management environment, including ticket management, assets, users, and administrative functions, confirming that the platform was correctly deployed and operational.

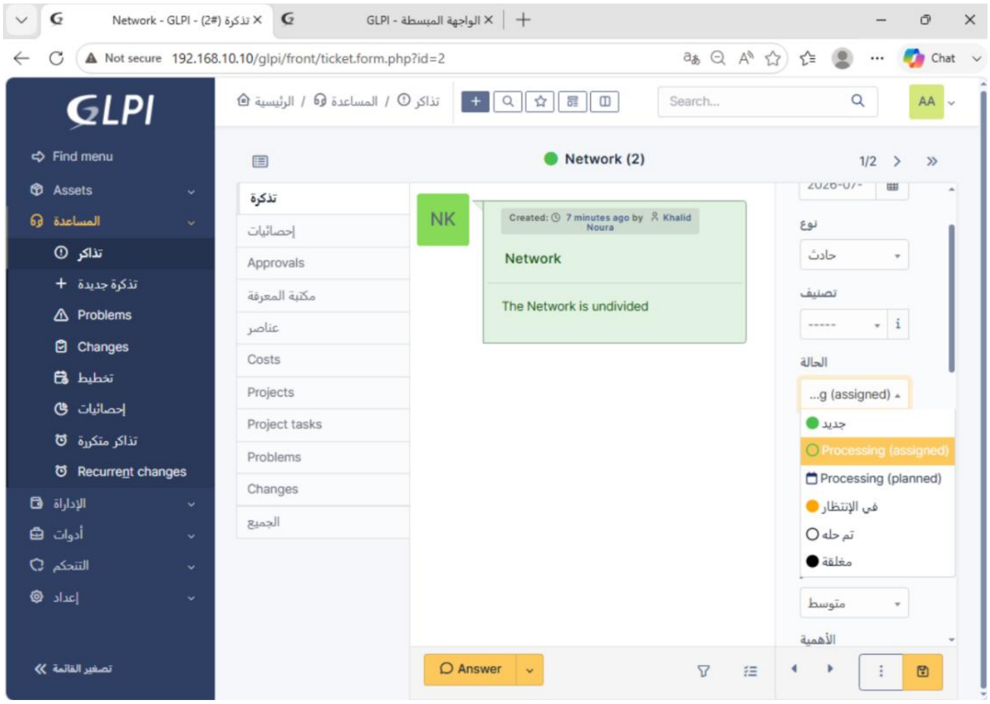
Ticket Creation and Queue Registration

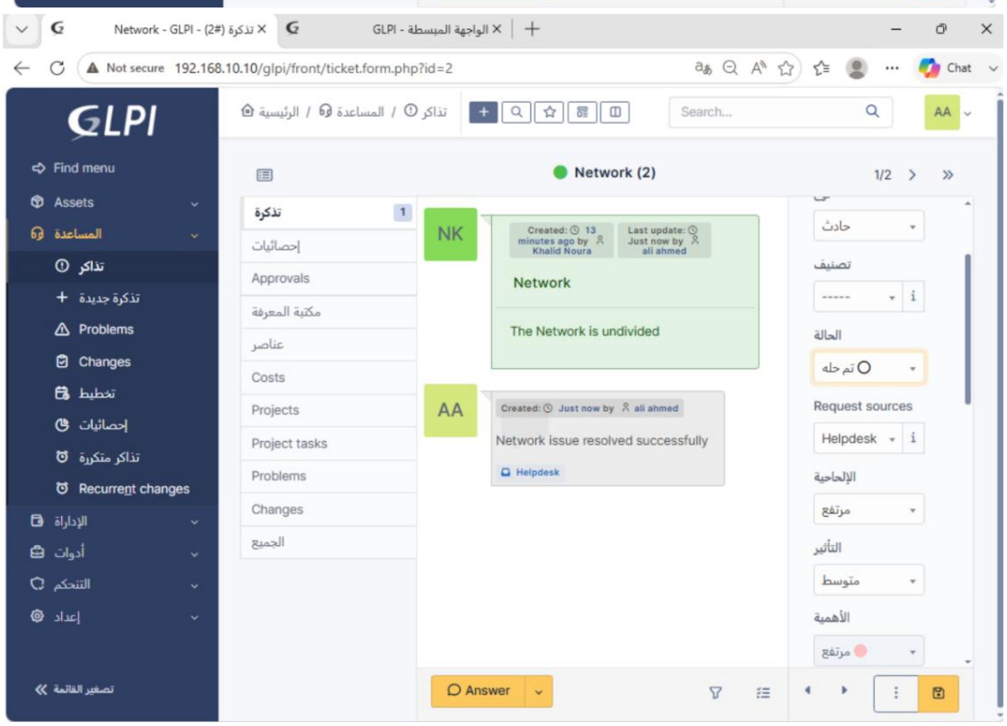




A new support ticket was successfully submitted through the GLPI helpdesk interface with the required incident details. After submission, the ticket was automatically registered in the IT support queue, confirming successful ticket creation, centralized incident logging, and proper routing within the IT Service Management (ITSM) workflow.

Ticket Assignment and Resolution





The support ticket was successfully assigned to an IT technician for investigation and resolution. After the troubleshooting process was completed, the ticket was marked as resolved within GLPI, demonstrating a complete end-to-end IT Service Management (ITSM) workflow from ticket assignment to successful incident resolution while maintaining centralized tracking and lifecycle management.

Final Environment Summary

The enterprise lab environment was successfully designed, implemented, and validated. The project integrated **Windows Server infrastructure**, Active Directory Domain Services (AD DS), DNS, Domain Join, Group Policy Objects (GPOs), PowerShell automation, and the GLPI IT Service Management (ITSM) platform into a single operational environment. The completed infrastructure demonstrates centralized identity management, security policy enforcement, automated administration, and an end-to-end IT support workflow, providing practical hands-on experience with technologies commonly used in enterprise IT environments.